PRIME MINISTERS' OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT MANYARA REGION
FORM FOUR MOCK EXAMININATION 2026

CHEMISTRY

TIME. 3:00HRS **AUGUST, 2026**

INSTRUCTIONS

1. This paper consists of three sections A, B, and C with total of eleven (11) questions.
2. Answer all questions from section A and B, and only 2 question from section C.
3. Section A carries 16 marks, section B carries 54 marks and section C carries 30 marks.
4. All writing should be in blue or black ink except for drawings which should be in pencil.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your **Examination number (index number)** on every page of your answers booklets provided

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SECTION A (16 Marks)

Answer all questions from this section

1. For each of the item (i-ix) choose the correct answer from the given alternatives and write it's latter beside the item number in the answer sheet provided.

i) Water can pass from the atmosphere to land and back into the atmosphere. What terms represents such a sequence?

A. Water purifications
B. Sedimentation
C. Electrolysis
D. Water cycle

ii) Which of the following gases if mixed with hydrogen would produce a very hot flame of up to $3000°C$

A. Oxygen
B. Neon
C. Chlorine
D. Argon

iii) Which of these statement is true about isotopes of an element?

A. The number of proton is the same, but the number of neutrons is different
B. The number of neutrons is the same, but the number of protons is different
C. The number of proton and neutrons is the same, but the number of electrons is different
D. The sum of proton and electrons

iv) Suffocations is a conditions in which the lungs

A. Do not exhale enough oxygen
B. Do not inhale enough oxygen
C. Do not inhale enough carbondioxide
D. Do not exhale enough carbondioxide

v) A form one student saw the "farmable" signs box and made the following possible interpretations which one is the most corrects?
A. The box contained fire wood
B. The box contained papers
C. The box had radioactive materials
D. The box contained spirits used in lung.

vi) When a small amount of common salts is dissolved in water the mixture so formed is
A. Homogeneous
B. Immiscible
C. Heterogeneous
D. Suspensions

vii) Identify the rock that does not cause hardness of water among the following
A. Dolomite
B. Gypsum
C. Limestone
D. Bauxite

viii) Which of the following condition of water is easily removal by boiling
A. Temporary hardness
B. Scum
C. Lime scale
D. Permanent hardness

ix) Which gas is produced when dilute nitric acid reacts with magnesium metals?
A. Nitrogen
B. Hydrogen
C. Oxygen
D. Nitrogen dioxide

x) Which of the following is true when a mixture of Bromine and methane is left in darkness?
A. Ethane and Hydrogen bromide are formed
B. Hydrogen bromide and bromomethane are formed
C. There is no visible reactions
D. Bromine is decolourised

2. Match the following terms in List A with the correct answer in list B.

LIST A	LIST B
i) Clinical thermometer ii) Suffocation iii) Bruises iv) Evaporating dish v) Crucible vi) Pipette	A. The shallow bowl with a curled lip commonly made of porcelain B. The skin injury that causes a change in colour of the skin C. The condition in which the lung are not getting enough oxygen causing difficult in breathing D. The container in which substance can be heated to very high temperature

	E. Titration chemicals. F. Used for measuring exact volume of liquid G. Used to hold solution

SECTION B (54 Marks)

Answer all questions in this section

3. A chemistry students of form four (IV) tested five solution A,B,C, D and E with a universal indicator solutions to determine their P^H value
   The results obtained are shown in the following table.

Beaker of solution	A	B	C	D	E
P^H	1	5	7	9	14

   Which solution was a
   i) Strong alkali?
   ii) Strong acid?
   iii) Weak acid?
   iv) Neutral solution?
   v) Weak base?

   (b) During neutralization reactions a students placed $25cm^3$ of sodium hydroxide in a flask and added few drops of phenolphthalein. The base required $22cm^3$ of dilute hydrochloric acid for complete neutralizations.
   a) What apparatus should have been used to measure the exactly volume of sodium hydroxide solution (01 Marks)
   b) What was the colour of the solution at the end of neutralization reaction? (01 Marks)
   c) What reagent was more concentrated than the other? Explain (02 Marks)
   d) Name the salt that was formed (0½ Marks)
   e) How would you obtain pure crystals of the salts resulting from this solution? (02 Marks)

4. Calculate the number of mole of copper (II) oxide reacted with Ammonia gas at STP and the volume of ammonia gas used if 20g of copper (II) oxide were reacted with Ammonia gas.

$$\underset{(s)}{3CUO} + \underset{(g)}{2NH_3} \longrightarrow \underset{(s)}{3CU} + \underset{(g)}{N_2} + \underset{(g)}{3H_2O}$$

5. (a) Give the common impurities found in metal ores
   (b) Why should these impurities be removed before the reactions process?
   (c) A student wishes to monitor the rate of reactions

$$\underset{(s)}{CaCO_3} + \underset{(aq)}{HCl} \longrightarrow \underset{(aq)}{CaCl_2} + \underset{(g)}{CO_2} + \underset{(l)}{H_2O}$$

Identify four (4) properties (other than the mass of calcium carbonate) that could be used to monitor the reactions rate and describes the changes that would occur for each.

6. (a) How many Faradays of electricity must be passed to deposit the mass of metals? 2.16g of silver

   (b) (i) If re – plate a chromium spoon, would you make it the anode or the cathode? Explain your answer given, the half reaction chemical equation and diagram.

   (ii) How many mole of electron are needed to plate out 0.49g of calcium (III)? If molar mass of chromium is cr =51.99.

7. (a) Give three examples of amphoteric hydroxide

   (b) Write equations to show the reactions of each of these hydroxide with dilute nitric acid and with excess sodium hydroxide solution

   (c) Study the reaction below and answer the questions

$$3KOH + FeCl_3 \longrightarrow 3KCl + Fe(OH)_3$$
$$\text{(aq)} \quad \text{(aq)} \quad \text{(aq)} \quad \text{(s)}$$

   i) What type of reaction is this?

   ii) Write the net ionic equation for the reaction.

8. Compound A ($C_4H_{10}O$) is oxidized with acidified potassium dichromate to give compound B. When it is dehydrated with conc. H_2SO_4, compound A gives compound C, while it undergoes hydrogenation to give compound D. Compound D is capable of undergoing a substitution reaction.
   Identify compound B, C, D and write their structures.

SECTION C (30 Marks)

Answer only two questions from this section

9. A doctor prescribed antacid tablets made from magnesium hydroxide to a student who was suffering from excess stomach acid.
   a) Why did the doctor prescribe the antacid tablets? (3 marks)
   b) Write a balanced chemical equation that takes place in the student's stomach after taking the tablet and the ionic equation. (3 marks)
   c) If the normal acid content in the stomach is 160.00 millimolar and the stomach secretes 0.5 litre of gastric juice in a day, calculate the mass of the tablet to be given to the student if the gastric juice had 210.00 millimolar hydrochloric acid.
   d) If each tablet of antacid weighs 145 mg of magnesium hydroxide, how many tablets will the student be given?
   e) If a dose requires 1 tablet every 12 hours, how many days should be the student take to finish the full dose?

10. Global warming is the increase of the average temperature on the surface of the earth as a result of enhanced greenhouse effects which is attributed to the

increase in the level of greenhouse gases in the atmosphere. State clearly the outcomes of the increase of greenhouse gases.

11. Your community is switching from using firewood to a modern fuel for cooking and heating to reduce indoor air pollution and deforestation. Explain six important factors you would consider when choosing the most suitable alternative fuel for widespread adoption in your community.